

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2012

PY 101 Engineering Physics

Date: 12.12.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct answer for the following questions.

[07]

- (a) The answer to the calculation $(3.5264 \times 0.00300 \times 225) / 72.215411$ should have how many significant figures?
- (i) 1 (ii) 2
(iii) 3 (iv) 4
- (b) If the percentage error in value of density is 2%, how should the true density can be reported?
- (i) 12.5 (ii) 0.05
(iii) 0.08 (iv) 1.25
- (c) If the time period of a vibrating body is 0.05 s, the frequency of waves it emits is _.
- (i) 5Hz (ii) 20Hz
(iii) 2Hz (iv) 200Hz
- (d) If the intensity of a source of sound is 10^{-7} W/m^2 , calculate the intensity level in dB.
- (i) 50 dB (ii) 60 dB
(iii) 80 dB (iv) 120 dB
- (e) A girl claps and hears the echo after reflection from cliff which is 660 m away. If the velocity of sound is 330 m/s, calculate the time taken for hearing the echo.
- (i) 2s (ii) 4s
(iii) 6s (iv) 10s
- (f) Sound and light waves both;
- (i) have similar wavelength (ii) obey the laws of reflection
(iii) travel as longitudinal waves (iv) travel through vacuum

(g) Which is the principle behind the construction of hologram?

- (i) Diffraction
- (ii) Absorption
- (iii) Interference
- (iv) Reflection

Q - 2 (a) In successive measurements for the width of a table, the readings turn out to be 31.33 cm, 31.15 cm, 31.26 cm, 31.02 cm and 31.20 cm. Calculate average deviation, standard deviation and standard error. [3]

(b) Define reverberation. The volume of the room is 600m^3 , the wall area of the room is 220 m^2 the floor area is 120 m^2 and ceiling area is 120 m^2 . The average sound absorption coefficient for wall is $0.03S$, for ceiling is $0.8S$ and for floor it is $0.06S$. Calculate reverberation time. [4]

(c) Explain why the reverberation time always has to be maintained at its optimum value. Hence discuss the important conditions to be satisfied for good acoustics. [7]

OR

Q - 2 (a) Consider an experiment to measure the velocity of ultrasound by using ultrasonic interferometer. What are the possible sources of systematic errors in this experiment? [3]

(b) Define relative intensity. What is the resultant sound level when a 70dB sound is added to an 80dB sound? [4]

(c) What is an acoustic grating? Explain the determination of ultrasonic wavelength using acoustic grating method. [7]

Q - 3 (a) Define (i) population inversion, (ii) pumping and (iii) optical resonator. [3]

(b) Is cladding necessary in a fiber cable? Explain. Core of an optical fiber has refractive index 1.545 and its cladding has refractive index 1.495. Calculate the critical angle, acceptance angle and numerical aperture. [4]

(c) What are Einstein's coefficients? Establish the relation between Einstein's coefficients. [7]

OR

Q - 3 (a) State three military applications of laser. [3]

(b) Discuss four advantageous of optical fiber communication system over the conventional coaxial communication system. [4]

(c) What is holography and hologram? Explain the method of construction & reconstruction of hologram. Give two applications of holography. [7]

SECTION – II

Q - 4 Choose the correct answer for the following questions.

[07]

- (a) A superconductor is a perfect _____ material.
(i) Diamagnetic (ii) Paramagnetic
(iii) Dielectric (iv) Insulator
- (b) The R_H of a specimen is $3.66 \times 10^{-4} m^3/C$. What is the mobility of carriers if its resistivity is $8.93 \times 10^{-3} \Omega m$.
(i) $0.04 m^2/Vs$ (ii) $0.04 cm^2/Vs$
(iii) $0.4 m^3/Vs$ (iv) $0.04 m^3/Vs$
- (c) The number of Bravais lattice in cubic crystal system is _____.
(i) 1 (ii) 2
(iii) 3 (iv) 4
- (d) Monochromatic X-rays of $\lambda = 1.5 \text{ \AA}$ are incident on a crystal face having an interplanar spacing of $1.6 \text{ \AA}$. What is the highest order for which Bragg's reflection can be seen?
(i) 1 (ii) 2
(iii) 3 (iv) 4
- (e) The packing factor of BCC structure is _____.
(i) 68 (ii) 52
(iii) 74 (iv) 47
- (f) Triclinic crystal system is represented by;
(i) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ$ (ii) $a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$
(iii) $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$ (iv) $a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
- (g) Miller indices of the plane parallel to X and Y axes are:
(i) (001) (ii) (010)
(iii) (100) (iv) (111)

Q - 5 (a) Define critical temperature, critical magnetic field and persistent current. [3]

(b) What is a Josephson junction? Distinguish between a.c. Josephson and d.c. Josephson effect. [4]

(c) What is Hall effect? Derive an expression for Hall voltage, Hall coefficient, and Hall mobility. [7]

OR

Q - 5 (a) Classify the solids based on the energy band model. [3]

(b) If a current of $10 mA$ flows through an X-ray tube at an operating potential of [4]

62000 V, what is the number of electrons striking the target per second and the minimum wavelength of the X-rays emitted.

(c) What are the characteristics of X-rays? How X-rays are generated? [7]

Q - 6 (a) Distinguish between crystalline & non-crystalline materials. [3]

(b) Explain diffraction in X-rays. Write Bragg's equation and hence define Bragg's law. [4]

(c) What are nanomaterials? What are the characteristic properties of objects in the nanoscale? [7]

OR

Q - 6 (a) Define crystal lattice, Bravais lattice and Miller indices. [3]

(b) Explain the unit cell and lattice parameters. What are primitive cell and a non-primitive cell? [4]

(c) How nanotechnology can be used for energy conservation, information & communication and heavy industry. [7]
